

Volcano Quest

Complete Tools & Packages Reference • Flutter Android • Target: July 2026

Core — must have

Recommended — strongly advised

Optional — enhances quality

1 — Flutter Core Packages (pub.dev)

Tool / Package	Type	Purpose	Used In
flutter_riverpod	State Mgmt	App-wide state management — XP, badges, progress, achievements	All screens
go_router	Navigation	Declarative routing between all game screens & levels	All screens
hive	Storage	Offline key-value DB for gameplay data & progress	Phase 2
hive_flutter	Storage	Flutter adapter for Hive — required alongside hive	Phase 2
firebase_core	Backend	Required Firebase initialisation package	Phase 2
cloud_firestore	Backend	Leaderboard & dashboard cloud sync	Phase 2
firebase_analytics	Analytics	Track user engagement and level completion events	Phase 2
lottie	Animation	Looping scene animations — lava, smoke, clouds	Phase 3-4
rive	Animation	Interactive character/UI animations with state machines	Phase 3-4
just_audio	Audio	Background music, SFX, and quiz feedback sounds	Phase 2
flutter_animate	Animation	Simple bounce/float motion for clouds & UI elements	Phase 3
confetti	FX	Celebration particle burst on level completion	Phase 3-4
model_viewer_plus	3D	Render GLTF/GLB 3D volcano model in Level 5	Level 5
flame	Particles	Spark, ember & smoke particle systems for background effects	Phase 3

* flame is used only for its particle system — you do not need the full game engine loop.

2 — Design & UI Tools

Tool / Package	Type	Purpose	Used In
Figma	Design	UI/UX mockups, screen flows, component library	Phase 1
LottieFiles.com	Animation	Download/preview free Lottie JSON animations	Phase 1-3
Rive.app	Animation	Create interactive Rive animations with state machines	Phase 3
Material Theme Builder	Design	Generate Flutter colour scheme from brand colours	Phase 1
FlutterFlow (opt.)	Prototyping	Rapid UI prototyping — optional visual aid	Phase 1

3 — 3D & Asset Creation Tools

Tool / Package	Type	Purpose	Used In
Blender	3D Modelling	Create/edit volcano 3D model, export to GLTF/GLB	Level 5
Meshy AI	AI 3D Gen	AI-generate 3D models from text prompts	Level 5
Sketchfab	Asset Source	Download free GLTF/GLB volcano & geology models	Level 5
Poly Pizza	Asset Source	Free low-poly 3D assets optimised for mobile	Level 5
CGTrader	Asset Source	Commercial 3D assets if free sources insufficient	Level 5
GLTF Viewer	3D Preview	Browser-based preview of GLTF/GLB before importing	Level 5
glTF Transform	Optimisation	CLI tool to compress & optimise GLB file size	Phase 4

* All 3D models should be exported as .glb (binary GLTF) and kept under 5MB for low-end Android performance.

4 — Audio Tools & Sources

Tool / Package	Type	Purpose	Used In
Freesound.org	SFX Source	Free CC-licensed sound effects — lava, quiz, ambient	Phase 2-4
Pixabay Audio	Music Source	Royalty-free background music tracks	Phase 2-4
Audacity	Audio Editor	Trim, normalise, and export audio to MP3/OGG	Phase 2
just_audio	Flutter Pkg	Playback engine — already listed in Section 1	Phase 2
audio_session	Flutter Pkg	Manage audio focus & interruptions on Android	Phase 2

5 — Firebase Services (console.firebase.google.com)

Tool / Package	Type	Purpose	Used In
Firebase Console	Platform	Project setup, rules, and service management	Phase 1
Cloud Firestore	Database	Leaderboard scores and user dashboard data	Phase 2
Firebase Analytics	Analytics	Level completion rates, session length, drop-off	Phase 2
Firebase Crashlytics	Monitoring	Real-time crash reports from test devices	Phase 5
Firebase App Distribution	Testing	Distribute test APKs to devices without Play Store	Phase 5

6 — Development, Build & Testing Tools

Tool / Package	Type	Purpose	Used In
Flutter SDK	Framework	Core framework — keep updated to stable channel	All phases
Android Studio	IDE	Primary IDE with Flutter/Dart plugin	All phases

Tool / Package	Type	Purpose	Used In
VS Code	IDE	Lightweight alternative with Flutter extension	All phases
GitHub	Version Ctrl	Source control, branching, and backup	Phase 1+
Android Emulator	Testing	Built-in emulator for rapid iteration	Phase 3-5
Physical low-end device	Testing	Real device test — essential for performance	Phase 3-5
flutter_test	Testing	Unit & widget tests for quiz engine & XP logic	Phase 5
flutter_driver	Testing	Integration/flow tests across screens	Phase 5
Dart DevTools	Profiling	Memory, CPU, and frame rate profiling	Phase 4-5
flutter build apk	Build	Generate release APK for submission	Phase 6

7 — Documentation & Presentation Tools

Tool / Package	Type	Purpose	Used In
Google Docs / Word	Docs	Thesis / capstone written documentation	Phase 6
Notion	Notes	Dev notes, task tracking, sprint planning	All phases
Figma	Slides	Demo presentation slides and screenshots	Phase 6
scrcpy	Screen Mirror	Mirror physical Android device to PC for demo	Phase 6
OBS Studio	Recording	Record gameplay demo video for presentation	Phase 6
Canva (opt.)	Design	Quick poster or presentation visuals	Phase 6

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